

INSTRUCTOR NOTES

A/B Testing & Randomized Experiments

ECON 4370 / BANA 4373 — Applied Data Tools for Economics & Business

Sam Houston State University | Spring 2026

Two-class block	One week (back-to-back sessions)
Class 1 total time	~75 minutes
Class 2 total time	~75 minutes
Companion notebook	ECON4370_AB_Testing.ipynb
Prerequisites	Potential outcomes (ATE, ATT, selection bias), DiD, OLS regression
Key new content	Power analysis, ITT vs LATE, multiple testing, attrition bias

These notes are written as a detailed teaching script and reference. The goal is to give you enough depth that you can field any question a sharp business student might ask, even after some time away from the material. Every section includes what to say, what to write on the board, common student confusions, and exact answers to anticipated questions.

TABLE OF CONTENTS

CLASS 1

1. Conceptual Overview — Why A/B Testing Belongs Here
2. Three-Act Opening (Netflix, Amazon, Bing)
3. From Observational to Experimental — The Quick Callback
4. Why Randomization Works — The Math
5. Key Vocabulary and Estimands
6. Worked Example — Email Subject Line Test (4 slides)
7. Power Analysis and Sample Size
8. Balance Checks
9. Class 1 Notebook Exercise — Teaching Script

CLASS 2

10. Difference-in-Means = OLS
11. Non-compliance: ITT vs LATE
12. Multiple Testing and P-hacking
13. Attrition Bias
14. Statistical vs Economic Significance
15. Break the Estimator — Notebook Exercise
16. Bridge to Text and Unstructured Data

APPENDIX

- A. Full Mathematical Derivations
- B. Common Student Questions Master List
- C. Timing Guide

CLASS 1 — THE LOGIC OF RANDOMIZATION

1. Conceptual Overview — Why A/B Testing Belongs Here

■ Read before class — not a slide section

Students have already seen the fundamental problem of causal inference through two lenses: the **potential outcomes framework** (ATE, ATT, selection bias) and **difference-in-differences** (parallel trends, Card & Krueger, Medicaid expansion). Both of those were workarounds for the fact that we could not randomize. This lecture closes the loop by showing what happens when we *can* randomize.

The narrative arc for these two classes is: *DiD was our best tool when randomization was impossible. A/B testing is what we do when it is possible — and it is possible far more often in business than in policy settings.* This framing motivates why business students should care about a topic that might feel like "just statistics."

Tip: Do not re-teach potential outcomes from scratch. Students have seen it. A two-minute callback table on the slide is enough. Spend the saved time on the worked example and power analysis, which are genuinely new.

Learning objectives for the two-class block:

- Explain why randomization eliminates selection bias using potential outcomes notation
- Design a business A/B test: specify MDE, power, sample size before running
- Execute a two-sample t-test and interpret results statistically and economically
- Diagnose and correct for non-compliance using ITT and LATE
- Identify p-hacking and apply multiple testing corrections
- Recognize attrition bias and its directional effect

2. Three-Act Opening — Netflix, Amazon, Bing

■ ~15 minutes total. Netflix: 4 min. Amazon: 5 min. Bing: 4 min. Punchline: 2 min.

ACT 1 — Netflix (The Relatable One)

Before showing the slide, put up the poll slide with the two thumbnails. Give students 30 seconds to vote silently. This is important — you want them committed to a prediction before the reveal. If your classroom has a polling tool (Poll Everywhere, Mentimeter, show of hands), use it. The point is to create a moment of investment.

"Take a look at these two images. Same movie. Two different thumbnails. Vote on which one you think gets more clicks. Just go with your gut — Version A or Version B? Take 30 seconds."

After the vote: Reveal that Netflix does not guess — they test. Every thumbnail on the platform has been A/B tested. The key number to land: conversion to "press play" varies by **up to 30%** based on thumbnail alone. That is a massive effect for a single visual choice.

"Here is the thing — Netflix does not ask their design team which thumbnail is better. They show Version A to half their users and Version B to the other half, they measure who actually clicks play, and the data decides. The designers do not get a vote. Nobody gets a vote. The experiment gets a vote."

Key lesson to plant: Even simple visual choices have large, measurable effects — and human intuition systematically fails to predict which version wins.

Tip: If students ask how Netflix handles the fact that different users have different tastes: great question. They actually run personalized thumbnail selection — different users see different thumbnails for the same title based on their viewing history. This is beyond simple A/B testing into multi-armed bandits and contextual bandits. Park it for later or acknowledge it as a preview of more advanced content.

ACT 2 — Amazon (The Surprising One)

Show the TikZ mockup slide with the two checkout forms side by side. Let students read it for a few seconds before you say anything. The visual does the work — they can see the change is literally one word.

"Amazon noticed that people were abandoning their checkout flow at a very specific step. The step where you had to create an account before buying. The fix their team proposed was to change one button. "Register" became "Continue." That is it. One word. They also added a small line underneath: "You can create an account later." The result was 45 million additional customers completing purchases in the first year. Which worked out to about 300 million dollars."

Now ask the discussion question directly:

"Here is what I want you to think about for 60 seconds. Could you have predicted that? If someone had said to you — "We are going to change one word on a button" — would you have said: that is worth 300 million dollars? Talk to the person next to you."

After 60 seconds of pair discussion, cold-call two or three pairs. The typical answers are: (1) no, they would not have predicted it, (2) maybe they would have predicted the direction but not the magnitude, (3)

occasionally a student says yes. Push back gently on (3): "How would you have known the magnitude? What would your confidence interval have been?"

Key lesson to land:

"The reason this is surprising is not that the change is clever. It is that human intuition about user behavior is extremely unreliable. The experiment does not care about your intuition. It measures reality. That is a fundamentally different epistemology from how most business decisions get made."

Tip: The original article is "The \$300 Million Button" by Jared Spool (UIE, 2009). It is worth reading in full before class — it has additional detail about WHY the registration requirement caused abandonment (people forgot their passwords, they did not want to commit to a relationship, etc.) that you can use if students ask.

ACT 3 — Bing (The Payoff)

This is the most important of the three because it introduces the organizational lesson that connects to why randomized experiments matter in a business context. The Netflix example showed that intuition fails. The Amazon example showed the scale of the stakes. The Bing example shows that the experiment can override the organizational hierarchy.

"The Bing story. A junior engineer at Microsoft had an idea for how ad headlines were displayed in Bing search results. He brought it to leadership. They said: 'interesting, but we have bigger priorities right now.' So he ran the test himself, on his own time, as a side project. The result: 100 million dollars in additional annual revenue. From one headline change. Leadership had deprioritized it. The experiment proved everyone wrong."

Now ask the big question — this is the one that drives the rest of the lecture:

"What is the difference between a senior manager saying 'I think X is better' and a randomized experiment saying 'X generates \$100 million more revenue'? What makes one more trustworthy than the other? I will give you a hint: it has a specific technical name."

Let students propose answers. Guide them toward: **selection bias, confounding, randomization**. The manager's opinion is based on observational experience — they have seen things work in certain contexts, but those contexts might differ from the current situation in ways that matter. The experiment breaks that link.

PUNCHLINE SLIDE

"In each of these three cases — Netflix, Amazon, Bing — the experiment knew something the experts did not. The question for today is: why? What is it about randomization that makes it epistemically special? That is what we are going to dig into."

3. From Observational to Experimental — The Quick Callback

■ ~5 minutes. This is a transition, not new content.

The callback table on the slide (PO framework → DiD → A/B Testing) is designed to give students a map of where they are in the course. Do not spend more than 3-4 minutes here. The goal is orientation, not review.

"Look at this table. We spent two weeks on potential outcomes — learning what ATE and ATT mean, why selection bias exists, why we cannot just compare treated and untreated groups. Then we spent a week on DiD — a clever identification strategy that uses a control group and a time dimension to get around the selection problem. What DiD and all observational methods have in common is that they are workarounds. They are what we do when randomization is not available. Today we ask: what happens when randomization IS available?"

The key conceptual move: In policy settings (Medicaid expansion, minimum wage), governments do not run randomized experiments — they make policy changes for political reasons, and researchers work backward to estimate effects. In business settings, randomization is cheap, fast, and legal. This is why A/B testing is the dominant causal inference tool in industry.

Tip: A student might ask: "If A/B testing is so great, why did we spend two weeks on observational methods?" Good question. Two reasons: (1) Not everything can be randomized — you cannot randomize whether someone gets cancer, whether a country changes its minimum wage, whether a firm goes bankrupt. (2) Observational methods are used to analyze data from experiments that broke (non-compliance, attrition) as well as from historical data. The tools are complementary.

4. Why Randomization Works — The Math

■ ~8 minutes. This is the conceptual core of Class 1.

This is the slide with the most formal math in Class 1. Students have seen potential outcomes notation before, so the notation should not be new. The goal is to show explicitly how randomization eliminates selection bias.

THE SELECTION BIAS DECOMPOSITION — FULL DERIVATION

Start by writing this on the board (or pointing to the slide). Walk through each step slowly.

$$\text{Naive estimate} = E[Y_i | D_i=1] - E[Y_i | D_i=0]$$

Add and subtract $E[Y_i^0 | D_i=1]$:

$$= E[Y_i^1 | D_i=1] - E[Y_i^0 | D_i=1] + E[Y_i^0 | D_i=1] - E[Y_i^0 | D_i=0]$$

$$= \text{ATT} + \text{Selection Bias}$$

The **ATT** is the treatment effect for those who actually received treatment. The **selection bias** term $E[Y_i^0 | D_i=1] - E[Y_i^0 | D_i=0]$ is the difference in untreated potential outcomes between treated and control groups — i.e., how much the two groups would have differed even without treatment.

"The selection bias term is the problem. It says: even if I give you no treatment at all, the people who would have chosen to be in the treatment group are fundamentally different from the people who would have stayed in the control group. In an email campaign, the people who click a link are already higher-intent than those who do not. We cannot observe this difference — it lives in the counterfactual."

Now show how randomization solves this:

Randomization: D_i is independent of (Y_i^0, Y_i^1)

Written: $D_i \text{ indep } (Y_i^0, Y_i^1)$

Therefore: $E[Y_i^0 | D_i=1] = E[Y_i^0 | D_i=0]$

Selection bias = 0

Naive estimate = ATT = ATE (since treatment is randomly assigned)

"When we randomize, we break the link between who gets treated and what their potential outcomes would have been. In expectation, the two groups are identical before treatment. So the only thing that can cause them to differ after treatment is the treatment itself. This is why the experiment beats the expert."

Tip: Students sometimes ask: "Does this mean ATE equals ATT in an experiment?" Yes — under complete randomization, $ATE = ATT = ATC$ because each subgroup is a random sample of the same population. This would not be true with stratified or targeted assignment.

Watch out: Some students will conflate "random assignment" with "random sampling." These are different. Random sampling is about how you draw your study population. Random assignment is about how you allocate units to treatment within that population. An experiment can have random assignment but a non-representative sample (e.g., only existing customers), and external validity is then a separate question.

5. Key Vocabulary and Estimands

■ ~5 minutes.

This slide is definitional. Move through it at a steady pace — students do not need to copy everything, it will be in the notebook. The goal is to give them vocabulary for the worked example that follows.

Assignment mechanisms — what to say:

- **Complete randomization:** Each unit independently assigned with fixed probability (e.g., 50%). This is the simple case. Both groups expected to be equal on all pre-treatment characteristics.
- **Stratified randomization:** Divide population into strata (e.g., by geography or prior purchase history) and randomize within each stratum. Guarantees balance on the stratification variable, reduces variance of the estimator. Commonly used in large-scale business experiments where one covariate (e.g., platform: iOS vs Android) is known to matter a lot.
- **Cluster randomization:** Assign entire groups (stores, schools, cities) rather than individuals. Used when treatment spillovers are likely — if you randomize at the individual level in a social network, control units might be affected by treated users they know. Reduces effective sample size; requires cluster-robust standard errors.

Key estimand to define clearly:

$$ATE = E[Y_i^1 - Y_i^0] = E[Y_i^1] - E[Y_i^0]$$

Under randomization, the difference in observed group means is an unbiased estimator of ATE:

$$ATE_{\hat{}} = \bar{Y}_{\text{treatment}} - \bar{Y}_{\text{control}}$$

Tip: Students sometimes ask why we use $E[Y^1] - E[Y^0]$ rather than $E[Y^1 - Y^0]$. These are the same by linearity of expectation: $E[Y^1 - Y^0] = E[Y^1] - E[Y^0]$. The second form is more tractable because we can estimate $E[Y^1]$ and $E[Y^0]$ separately from observed group means.

6. Worked Example — Email Subject Line Test (4 slides)

■ ~20 minutes total. Setup: 4 min. Data: 4 min. Test: 7 min. Interpretation: 5 min.

This is the technical heart of Class 1. Students need to see a complete end-to-end example with real numbers before they touch the notebook. Walk through every step on the slide, narrating each calculation.

SLIDE 1 — THE SETUP

"Here is our scenario. An online retailer wants to know whether personalizing the email subject line increases revenue. They have two versions: a generic one — 'Shop our new arrivals' — and a personalized one — 'Maria, your picks are in.' The outcome is revenue per customer in the seven days after the email is sent. They have 10,000 customers and split them 50-50 at random."

Before showing the data slide, invoke the prior:

"Before I show you the results — write down a number. How much more revenue do you think the personalized email generates, per customer, compared to the generic one? Write an actual dollar amount. I will give you 30 seconds."

This "write your prior" exercise is deliberate. Students who commit to a number before seeing data learn more from the reveal than those who do not. It also generates discussion: "I said \$5, you said \$0.50 — what drove that difference?"

SLIDE 2 — THE DATA

Summary statistics (what the numbers mean):

- Mean revenue Control (A): \$12.44 per customer
- Mean revenue Treatment (B): \$14.11 per customer
- Raw difference: \$1.67 — this is our estimated ATE
- Standard deviation in both groups: ~\$18 — this is large relative to the effect, which is why the t-test is necessary

The distribution sketch: Point out that the two distributions overlap almost completely. The means are only \$1.67 apart but the standard deviation is \$18. This is typical of revenue data — most customers buy nothing (zero revenue), a few buy a lot (heavy right tail). The overlap is why we cannot just "look at the data" and declare a winner — we need formal inference.

"This is the picture that should make you nervous. The treatment mean is higher, yes. But these distributions are almost identical. The \$1.67 gap could easily be random noise. That is what the t-test is going to tell us — is this gap larger than what we would expect to see by chance alone?"

The balance check table: Briefly note that the pre-treatment covariates (age, prior purchases, account age) are nearly identical across groups. This is evidence that randomization worked — the groups are comparable before treatment.

"Before I even look at the outcome, I always check balance. If the treatment group had systematically older customers, or customers who had bought more in the past, the comparison would be unfair. Look at this table — the numbers are almost identical. That is what good randomization looks like."

SLIDE 3 — THE TEST

Walk through the t-test calculation step by step on the slide. Write each line on the board as you say it.

Step 1 — State the hypotheses:

$H_0: \mu_B - \mu_A = 0$ (no effect)

$H_1: \mu_B - \mu_A \neq 0$ (two-tailed test)

"We are testing whether the difference in population means is zero. Note that I am doing a two-tailed test — I am open to the possibility that the personalized email could backfire and reduce revenue. In practice, you might do a one-tailed test if you are only interested in improvement, but I recommend two-tailed as a default."

Step 2 — Compute the standard error:

$SE = \sqrt{s_A^2 / n_A + s_B^2 / n_B}$

$= \sqrt{18.20^2 / 5000 + 18.65^2 / 5000}$

$= \sqrt{331.24/5000 + 347.82/5000}$

$= \sqrt{0.06625 + 0.06956}$

$= \sqrt{0.13581}$

$= 0.3686 \approx 0.369$

The standard error measures how much we expect our estimate $\hat{\tau}$ to vary across different random samples of the same size. It decreases as n grows (bigger sample $\rightarrow$ more precise estimate) and increases as the variance of the outcome grows (noisier outcome $\rightarrow$ harder to detect effects).

Step 3 — Compute the t-statistic:

$t = \tau_{\text{hat}} / SE = 1.67 / 0.369 = 4.52$

"The t-statistic tells us how many standard errors our estimate is away from zero. A value of 4.52 means our observed difference of \$1.67 is 4.52 standard errors above zero. Is that surprising under the null hypothesis? The critical value for a two-tailed test at the 5% level is 1.96. Our t of 4.52 is way past that. So yes — this is very surprising under the null."

Explain the t-distribution diagram:

- The bell curve is the sampling distribution of t under H_0 (no effect)
- The red shaded regions are the rejection regions — if t falls there, we reject H_0
- The red dashed lines are at ± 1.96 — the critical values for $\alpha = 0.05$

- Our $t = 4.52$ lands far to the right, past the critical value
- $p\text{-value} < 0.001$: if there were truly no effect, the probability of seeing a t this extreme or more extreme is less than 0.1%

Watch out: Students sometimes confuse the p -value with the probability that H_0 is true. Be explicit: the p -value is $P(\text{seeing data this extreme} \mid H_0 \text{ is true})$. It is NOT $P(H_0 \text{ is true} \mid \text{data})$. That would require Bayesian methods.

SLIDE 4 — INTERPRETING THE RESULT

The confidence interval:

$$CI = \tau_{\text{hat}} \pm 1.96 * SE = 1.67 \pm 1.96 * 0.369 = 1.67 \pm 0.723$$

$$CI = [\$0.95, \$2.39]$$

"A 95% confidence interval means: if we repeated this experiment many times, 95% of the intervals we computed this way would contain the true parameter. This interval runs from \$0.95 to \$2.39. The whole thing is above zero, which confirms statistical significance. But more usefully — it gives us a plausible range for the true effect. Even the lower bound, \$0.95, is economically meaningful at scale."

The regression equivalence — write this on the board:

$$\text{Revenue}_i = 12.44 + 1.67 * \text{Treatment}_i + \epsilon_i$$

"Here is something elegant. The difference-in-means estimator is identical to the OLS coefficient on a binary treatment indicator. When you run this regression, the intercept is the control group mean, and the coefficient on Treatment is the difference in means — which is our ATE estimate. Same point estimate, same standard error, same p -value. This is what the notebook will produce when you call `statsmodels.OLS`."

The scale-up calculation:

$$\text{Monthly impact} = 500,000 \text{ emails/month} * \$1.67/\text{customer} = \$835,000/\text{month}$$

"This is economically significant. We are not just talking about p -values. A \$1.67 per-customer effect, multiplied across half a million emails per month, is 835,000 dollars per month, or about 10 million dollars per year. That is real money. Changing the subject line costs almost nothing to implement. This is the power of running experiments at scale."

Tip: Anticipated question: "What if the effect decays over time as users get used to personalized emails?" Great question. This is called novelty effect or Hawthorne effect. A well-run experiment should be long enough to capture steady-state behavior, not just the initial reaction. This is a real limitation of short-term A/B tests.

7. Power Analysis and Sample Size

■ ~10 minutes. This is genuinely new content — spend the time.

Power analysis is one of the most practically important topics in this lecture and also one of the most frequently skipped in standard statistics courses. Students need to understand it not just mathematically but intuitively — it determines whether their experiment is worth running at all.

THE FOUR COMPONENTS

1. Significance level alpha:

- The probability of a false positive (Type I error) — rejecting H_0 when it is actually true
- Standard default: $\alpha = 0.05$ (5% false positive rate)
- High-stakes decisions (medical trials, large financial commitments) may use $\alpha = 0.01$
- Industry A/B testing platforms sometimes use $\alpha = 0.10$ to run faster experiments

2. Power (1 - beta):

- The probability of detecting an effect when one truly exists (avoiding Type II error)
- Standard default: 80% power ($\beta = 0.20$)
- Higher-stakes: 90% power
- Intuition: if the true effect is δ , you will detect it 80% of the time with 80% power

3. Minimum Detectable Effect (MDE):

- The smallest effect size you would consider practically meaningful
- This is a business decision, not a statistical one
- Example: "We do not care about improving click-through rate by less than 0.5 percentage points"
- Smaller MDE = larger required sample. This is the key tradeoff.

4. Sample size n (derived from the above three):

$$n \geq (z_{1-\alpha/2} + z_{1-\beta})^2 \cdot 2\sigma^2 / \delta^2$$

$$\text{Common defaults: } z_{0.975} = 1.96, z_{0.80} = 0.84$$

$$\text{So: } n \geq (1.96 + 0.84)^2 \cdot 2\sigma^2 / \delta^2 = (2.80)^2 \cdot 2\sigma^2 / \delta^2$$

$$= 7.84 \cdot 2\sigma^2 / \delta^2 = 15.68 \cdot \sigma^2 / \delta^2$$

"Let me show you what this formula tells us intuitively. The sample size grows with sigma-squared — noisier outcome data requires bigger samples. The sample size shrinks with delta-squared — if you only care about big effects, you need fewer observations. This is the fundamental tradeoff in experiment design."

Worked numerical example for the email experiment:

$$\text{Suppose: } \delta = \$1.50 \text{ (MDE), } \sigma = \$18, \alpha = 0.05, \text{ power} = 0.80$$

$$n = 15.68 * (18)^2 / (1.50)^2$$

$$= 15.68 * 324 / 2.25$$

$$= 5,080 / 2.25$$

$$= 2,258 \text{ per group} \Rightarrow \sim 4,500 \text{ total}$$

"So with 5,000 per group — which is what we had — we were well-powered to detect an effect of \$1.50 or larger. Our actual effect was \$1.67, which is above the MDE, so we would expect to detect it with high probability. This is reassuring."

Watch out: A critical practical point: the MDE calculation requires you to know sigma before you run the experiment. Where does sigma come from? Historical data on the same metric from prior periods. This is one reason why having good data infrastructure is a prerequisite for running good experiments.

Tip: For the notebook exercise, students will use `scipy.stats.ttest_ind` and also compute power manually using the formula. A useful check: `scipy.stats.ttest_ind_from_stats` can verify the formula result. Also point them to `statsmodels.stats.power.tt_ind_solve_power` for the industrial version.

8. Balance Checks

■ ~5 minutes.

Balance checks are the first thing a researcher or data scientist does after assignment. They verify that randomization worked. If balance checks fail, the experiment may be compromised and results should be reported with caution.

What to check and how:

- Compare means of all pre-treatment covariates across treatment and control
- Use the standardized difference (also called Cohen's d for balance): $d = (\text{mean}_T - \text{mean}_C) / \text{pooled_SD}$
- Rule of thumb: $|d| < 0.1$ indicates good balance
- Do NOT use hypothesis tests (t-tests) for balance checks — with large n, even trivial imbalances become "significant." The standardized difference is scale-free and more appropriate.

What to do if balance fails:

- First: check the assignment code. Was the randomization truly implemented correctly?
- Second: if the imbalance is on a key covariate, include it as a control variable in the regression (ANCOVA) to absorb its variance
- Third: if imbalance is severe across many covariates, the experiment may be compromised. Report cautiously and consider re-running.

Watch out: In real business settings, assignment bugs are more common than students expect.

Cookie-based assignment can be compromised by users clearing cookies. Session-based assignment can cause the same user to appear in both groups. Always check balance — it catches implementation errors, not just bad luck.

9. Class 1 Notebook Exercise — Teaching Script

■ ~20 minutes. Students work individually first, then discuss.

The notebook has four parts. Walk students through the structure before they start. The "write your prior" cell must be completed before running any code.

SOLUTION KEY — Part 1: Clean RCT

The notebook simulates data with `TRUE_EFFECT = 2.0`. Expected outputs:

Solution key: The difference-in-means estimate should be approximately 2.0 (within sampling error, typically 1.7 to 2.3 with the default sample size). The t-statistic should be well above 1.96, and the p-value should be less than 0.05. The regression coefficient on Treatment should match the difference-in-means to within rounding error.

SOLUTION KEY — Part 2: Biased Assignment

The notebook modifies assignment so high-intent customers (higher baseline revenue) are more likely to receive treatment. Expected outputs:

Solution key: The biased estimate should be substantially higher than 2.0 — typically 3.5 to 5.0 depending on the strength of the selection mechanism. The balance check will show large standardized differences on the baseline revenue covariate ($d > 0.5$). This demonstrates upward bias from selection: treated customers would have had higher revenue even without treatment.

SOLUTION KEY — Part 3: Balance Check

Solution key: Under clean randomization: all standardized differences should be below 0.1. Under biased assignment: the `baseline_revenue` variable should show $d > 0.3$, and possibly other correlated variables too. The function `compute_smd(df, covariate, treatment_col)` should return a table with columns: `covariate`, `mean_control`, `mean_treatment`, `smd`.

Common student errors to watch for:

- Using `np.random.random()` without setting a seed — results will not be reproducible
- Confusing the treatment indicator column (0/1) with the treatment effect (a float)
- Forgetting to re-run the data generation cell after modifying the assignment mechanism
- Using the biased sample to compute balance statistics but the clean sample for the t-test (or vice versa)

CLASS 2 — WHERE EXPERIMENTS BREAK

10. Difference-in-Means = OLS

■ ~7 minutes. Brief but important for connecting to what they know.

Students have used OLS for several lectures. This section makes explicit that the simple A/B test estimator is a special case of OLS. This is important because it (a) reduces cognitive load — no new estimator to learn — and (b) immediately generalizes to the ANCOVA extension.

The algebraic identity:

Model: $Y_i = \alpha + \tau \cdot D_i + \epsilon_i$

OLS gives: $\tau_{\text{hat}} = \text{Cov}(Y_i, D_i) / \text{Var}(D_i)$

When D_i is binary with equal group sizes:

$\tau_{\text{hat}} = Y_{\text{bar}_1} - Y_{\text{bar}_0}$ (difference in group means)

"You have been running t-tests this whole time without knowing you were doing OLS. The coefficient on a binary indicator in a simple linear regression is exactly the difference in group means. Same estimate, same standard error, same p-value. This is not a coincidence — it is a theorem."

The ANCOVA extension:

$Y_i = \alpha + \tau \cdot D_i + \beta \cdot X_i + \epsilon_i$

Adding pre-treatment covariates X_i does not change the expected value of τ_{hat} (it is still unbiased under randomization), but it **reduces the residual variance**, which reduces the standard error of τ_{hat} . This means you can detect the same effect with a smaller sample, or detect smaller effects with the same sample. This is one of the main reasons to collect pre-treatment data before running an experiment.

Tip: Students sometimes worry that adding covariates will "change the result." Under randomization, adding covariates to the regression should not meaningfully change the point estimate of τ . If it does change substantially, that is a red flag indicating that randomization may not have worked (balance check failure).

11. Non-compliance: ITT vs LATE

■ ~15 minutes. This is the most technically demanding part of Class 2.

Non-compliance is where most real-world A/B tests differ from textbook experiments. Walk through the business motivation carefully before introducing the formal framework.

THE BUSINESS INTUITION

"Imagine you run a promotion experiment. You randomly select 5,000 customers to receive a discount coupon by email. The other 5,000 get no email. But not everyone opens their email. Maybe 40% of the treatment group never even sees the coupon. They were assigned to treatment but they did not receive treatment in any meaningful sense. What do you do?"

The wrong thing to do: Compare only coupon-openers to the control group.

"If I compare coupon-openers to the control group, I have reintroduced selection bias. Why? Because customers who open promotional emails are systematically different from those who do not. They are more engaged, more likely to buy anyway. I have broken my randomization."

COMPLIER TYPES

There are four types of units in an experiment with non-compliance:

Type	Definition	Business Example
Compliers	Take treatment when assigned; don't when not	Open coupon email if assigned; don't seek it out otherwise
Never-takers	Never take treatment regardless of assignment	Never open promotional emails, ever
Always-takers	Always take treatment regardless of assignment	Always find and use coupons even if not sent one
Defiers	Do the opposite of assignment (rare)	Only use coupon if NOT assigned (spite shoppers)

Under the **monotonicity assumption** (no defiers), we can identify two key estimands:

ITT — INTENTION TO TREAT

$$ITT = E[Y_i | Z_i=1] - E[Y_i | Z_i=0]$$

Where Z_i is the *assignment* indicator (not the treatment receipt indicator D_i). The ITT compares groups based on assignment, regardless of whether they actually received treatment. It is always unbiased because Z_i was randomized.

What ITT answers: "What is the effect of being *offered* the treatment?" This is often the policy-relevant question. If you are deciding whether to run an email campaign, you want to know the effect of sending the emails — including the fact that many people will not open them.

ITT is always conservative because it dilutes the effect across compliers and non-compliers.

LATE — LOCAL AVERAGE TREATMENT EFFECT

$$\text{LATE} = \text{ITT} / \text{Compliance Rate}$$

$$= [E(Y|Z=1) - E(Y|Z=0)] / [E(D|Z=1) - E(D|Z=0)]$$

This is the Wald estimator — the ratio of the reduced form (ITT) to the first stage (the effect of assignment on actual treatment receipt). Under monotonicity, this identifies the ATE **for compliers only** — hence "Local" ATE.

Example calculation:

ITT = \$0.80 (effect of sending the coupon email)

Compliance rate = 0.40 (40% of assigned customers actually opened the email)

LATE = \$0.80 / 0.40 = \$2.00 (effect of the coupon FOR those who opened it)

"The LATE says: for customers who actually engage with our emails when we send them — the compliers — the coupon increases revenue by \$2. That is the effect we care about for understanding the mechanism. But if we are running a policy decision about whether to send the email campaign, the ITT is more relevant."

Tip: LATE is estimated via Instrumental Variables (IV), where the assignment Z is used as an instrument for actual treatment D . The OLS second stage gives the LATE. In Python: statsmodels does not have a dedicated IV estimator in the simple API, but linearmodels.iv.IV2SLS handles this cleanly. For the notebook, students use the Wald estimator formula directly: $\text{LATE} = \text{ITT} / \text{first_stage}$.

Watch out: LATE only identifies the effect for compliers. The LATE from this experiment cannot be applied to never-takers (who never open emails) because we have no information about how they would respond to treatment if they had received it. This is the "local" in LATE — it is local to the complier subpopulation.

Q: Is the LATE always larger than the ITT?

A: Yes, if the compliance rate is less than 1 (which it always is in practice). $\text{LATE} = \text{ITT} / \text{compliance rate}$, and $0 < \text{compliance rate} < 1$, so $\text{LATE} > \text{ITT}$. The ITT underestimates the per-complier effect because it averages over non-compliers who received no treatment (and thus got no effect).

Q: What if some control group members somehow get the treatment (always-takers)?

A: Then the compliance rate calculation changes. The first stage is $E(D|Z=1) - E(D|Z=0)$, which accounts for both the never-takers in the treatment group and the always-takers in the control group. The Wald formula still works — you just need to measure actual treatment receipt (D) separately from assignment (Z) for everyone.

12. Multiple Testing and P-hacking

■ ~10 minutes.

This section is crucial for business students who will work in environments where "run tests until something is significant" is a common (and dangerous) practice.

THE CORE MATH

$P(\text{at least one false positive in } m \text{ independent tests at level } \alpha)$

$$= 1 - (1 - \alpha)^m$$

$$\text{For } m=20, \alpha=0.05: 1 - (0.95)^{20} = 1 - 0.358 = 0.642$$

"If you run 20 independent tests at the 5% level, and there is no true effect anywhere, you have a 64% chance of finding at least one "significant" result. That is not a discovery. That is a coin flip weighted toward false positives."

How p-hacking happens in practice:

- Peeking: checking results before the pre-determined end date and stopping when significant
- Metric shopping: measuring 15 outcomes, reporting only the 2 that are significant
- Subsetting: splitting data by demographic subgroups until one shows significance
- Outlier exclusion: removing data points that move the result the wrong way
- Stopping and re-running: running the same test multiple times with slight variations

Corrections:

- **Bonferroni correction:** Set $\alpha^* = \alpha / m$. Guarantees family-wise error rate (FWER) at level α . Very conservative — controls probability of ANY false positive, even one. Use when even one false positive is costly.
- **Benjamini-Hochberg (BH):** Controls False Discovery Rate (FDR) — the expected proportion of rejected null hypotheses that are false positives. Less conservative than Bonferroni. Algorithm: sort p-values, find largest k such that $p_{\{(k)\}} \leq k \cdot \alpha / m$, reject all null hypotheses with $p \leq p_{\{(k)\}}$.
- **Pre-registration:** Declare your primary hypothesis, sample size, and stopping rule before collecting data. Ties your hands against post-hoc manipulation. Standard in academic research; increasingly common in industry.

The p-curve explanation for the histogram slide:

"Under the null hypothesis — if there is truly no effect — p-values are uniformly distributed between 0 and 1. That means you would expect roughly the same number of p-values in every interval: [0, 0.05], [0.05, 0.10], [0.10, 0.15], and so on. When researchers p-hack, you see a suspicious spike just below 0.05 — because studies that got p-values of 0.04 get published, and studies that got 0.06 go in the file drawer. That spike is the statistical fingerprint of selective reporting. It is called the p-curve."

Tip: A great real-world reference: Uri Simonsohn, Leif Nelson, and Joseph Simmons published "p-curve: A Key to the File Drawer" in 2014 in Journal of Experimental Psychology. They developed tools to detect

p-hacking in published literatures. The p-curve app is publicly available. This is a great follow-up reading for students interested in research methods.

13. Attrition Bias

■ ~7 minutes.

Attrition is often overlooked but can badly distort experimental results. The key conceptual point: attrition is only a problem if it is *differential* — i.e., if dropout is related to the outcome or to potential outcomes.

The formal problem: If attrition is related to potential outcomes, the remaining sample is no longer a random subset of the original. Selection bias is reintroduced.

Diagnosis:

- Check: is the attrition rate the same in treatment and control? (Test: regress an indicator for "dropped out" on treatment assignment — should be insignificant)
- Check: among those who remain, are pre-treatment characteristics balanced?

Direction of bias:

- If high-outcome units drop out of control but not treatment: treatment group looks better than it is (upward bias)
- If low-engagement units drop out of treatment (because the new UX is confusing): remaining treatment group is high-engagement, comparison to full control group produces upward bias
- If users are more likely to drop out when they are frustrated (and treatment causes frustration): attrition reveals the mechanism and the bias direction is ambiguous

Remedies:

- Collect outcomes at a fixed endpoint regardless of engagement (e.g., 30-day revenue window)
- Bound the effect: compute Lee bounds (worst-case upper and lower bounds on the treatment effect under attrition)
- Impute missing outcomes under conservative assumptions

14. Statistical vs Economic Significance

■ ~5 minutes. Brief — but the key message is important.

This is particularly important for business students who will work in large-scale tech environments where sample sizes of millions make almost everything "significant."

"With $n = 10$ million users, a change that increases click-through rate by 0.001 percentage points will be statistically significant at $p < 0.001$. That does not mean you should ship it. The p-value tells you whether the effect is real. It does not tell you whether it matters. These are completely separate questions."

What to report in a business A/B test result:

- Point estimate and confidence interval (in the units of the outcome metric)
- Relative lift (% change from baseline) — executives understand percentages
- Scale-up: what does this effect mean at full deployment (annual revenue impact, etc.)
- Cost of implementation: what does it cost to make this change permanent?
- Statistical significance: p-value and confidence interval

The Bayesian framing for business decisions:

"Before you run the experiment, ask yourself: what effect size would actually change my business decision? If I would implement the change for a \$0.10 per-customer effect and also for a \$10 per-customer effect, then the MDE is \$0.10 and I need a large sample. If I only care about effects above \$5, I can run a much smaller experiment. This question — what would change my decision — is your MDE. And it should be the same as your economic significance threshold."

15. Break the Estimator — Notebook Exercise

■ ~25 minutes. Students work in pairs.

This is the core applied exercise for Class 2. Students systematically violate assumptions and observe the consequences. The expected results table on the slide gives them something concrete to aim for.

SOLUTION KEY — Violation 1: Non-compliance

Solution key: With 40% non-compliance (40% of treatment group does not receive treatment): ITT should be approximately $0.60 * \text{TRUE_EFFECT} = 0.60 * 2.0 = 1.20$ (diluted effect). $\text{LATE} = \text{ITT} / \text{compliance_rate} = 1.20 / 0.60 = 2.00$ (recovers true effect). The naive comparison (among actual takers only) will be biased upward because compliers are self-selected — they tend to be higher-intent customers.

SOLUTION KEY — Violation 2: P-hacking

Solution key: 100 tests on pure noise ($\text{TRUE_EFFECT} = 0.0$): expect approximately 5 false positives (5% of 100) at $\alpha = 0.05$. After Bonferroni correction ($\alpha^* = 0.05/100 = 0.0005$): expect approximately 0 false positives. The p-value histogram should look roughly uniform (not concentrated near 0), confirming there is no real effect.

Code hint for students who get stuck:

```
import numpy as np from scipy import stats np.random.seed(42) n_tests = 100
p_values = [] for _ in range(n_tests): a = np.random.normal(0, 1, 500) b =
np.random.normal(0, 1, 500) _, p = stats.ttest_ind(a, b) p_values.append(p)
false_positives = sum(p < 0.05 for p in p_values) bonf_threshold = 0.05 / n_tests
bonf_fp = sum(p < bonf_threshold for p in p_values)
```

SOLUTION KEY — Violation 3: Attrition

Solution key: Dropping the bottom 30% of potential outcomes from the control group: the remaining control group has a higher mean than the full population. The difference estimate (treatment - remaining control) will be smaller than the true effect — downward bias in this case (control mean is inflated). If instead high-outcome units drop from control: the control mean falls, and the estimated treatment effect is inflated upward. The direction depends on which end of the outcome distribution attrits.

Discussion questions for the debrief (last 5 minutes):

- Which failure mode produces the largest bias? (Usually attrition, if severe)
- In the non-compliance case, which estimate would you report in a business memo — ITT or LATE? Why?
- If you ran 100 tests in your company per year, how many false positives would you expect even with perfect experiments?

16. Bridge to Text and Unstructured Data

■ ~5 minutes. Closing and motivation for next unit.

"We have spent two weeks building the machinery for causal inference with structured data. But in the real world, a lot of what happens in an experiment is captured in unstructured form. When you run an A/B test on a new product feature, customers write reviews. They submit support tickets. They post on social media. How do you measure whether treatment changed what people are saying about you? How do you analyze open-ended survey responses from an experiment? That is what the next unit is about — and the same causal framework applies. The outcome variable just looks different."

APPENDIX A — FULL MATHEMATICAL DERIVATIONS

A1. Unbiasedness of the Difference-in-Means Under Randomization

$$\begin{aligned}
 &E[Y_{\text{bar}_1} - Y_{\text{bar}_0}] \\
 &= E[Y_{\text{bar}_1}] - E[Y_{\text{bar}_0}] \\
 &= E[Y_i \mid D_i=1] - E[Y_i \mid D_i=0] \\
 &= E[Y_i^1 \mid D_i=1] - E[Y_i^0 \mid D_i=0] \text{ (consistency/SUTVA)} \\
 &= E[Y_i^1 \mid D_i=1] - E[Y_i^0 \mid D_i=0] \\
 &= E[Y_i^1] - E[Y_i^0] \text{ (randomization: } D_i \text{ indep } Y_i^0, Y_i^1) \\
 &= \text{ATE}
 \end{aligned}$$

The key step is the independence condition from randomization. Under selection into treatment, $E[Y_i^1 \mid D_i=1] \neq E[Y_i^1]$ because compliers and non-compliers differ on potential outcomes. Randomization eliminates this.

A2. Variance of the Difference-in-Means Estimator

$$\begin{aligned}
 \text{Var}(Y_{\text{bar}_1} - Y_{\text{bar}_0}) &= \text{Var}(Y_{\text{bar}_1}) + \text{Var}(Y_{\text{bar}_0}) \text{ [independence of groups]} \\
 &= \sigma_1^2 / n_1 + \sigma_0^2 / n_0 \\
 \text{SE} &= \sqrt{\sigma_1^2 / n_1 + \sigma_0^2 / n_0}
 \end{aligned}$$

In practice, we substitute sample variances s_1^2 and s_0^2 for population variances. Under H_0 , some textbooks use a pooled variance estimate (assuming equal population variances). The Welch t-test (scipy default) does not pool — it is more robust and generally preferred.

A3. Power Derivation

We reject H_0 when $|t| > z_{\{1-\alpha/2\}}$. Power is the probability of rejecting H_0 when the true effect is δ .

$$\begin{aligned}
 \text{Power} &= P(|t| > z_{\{1-\alpha/2\}} \mid \delta \neq 0) \\
 &\sim P(t > z_{\{1-\alpha/2\}} \mid \delta \neq 0) \text{ [one-sided approximation for large } \delta] \\
 \text{Under } H_1: t &\sim N(\delta/\text{SE}, 1) \\
 \text{Power} &= P(N(0,1) > z_{\{1-\alpha/2\}} - \delta/\text{SE})
 \end{aligned}$$

$$= \Phi(\delta/\text{SE} - z_{\{1-\alpha/2\}})$$

$$\text{Set Power} = 1-\beta: \Phi(\delta/\text{SE} - z_{\{1-\alpha/2\}}) = 1-\beta$$

$$\Rightarrow \delta/\text{SE} - z_{\{1-\alpha/2\}} = z_{\{1-\beta\}}$$

$$\Rightarrow \delta / \sqrt{2\sigma^2/n} = z_{\{1-\alpha/2\}} + z_{\{1-\beta\}}$$

$$\Rightarrow n = (z_{\{1-\alpha/2\}} + z_{\{1-\beta\}})^2 * 2\sigma^2 / \delta^2$$

A4. The Wald Estimator for LATE

Let Z_i = assignment indicator, D_i = treatment receipt indicator

First stage (effect of assignment on receipt):

$$\pi_1 = E[D_i | Z_i=1] - E[D_i | Z_i=0] \text{ (compliance rate)}$$

Reduced form (ITT):

$$\rho = E[Y_i | Z_i=1] - E[Y_i | Z_i=0]$$

Wald (LATE):

$$\text{LATE} = \rho / \pi_1 = \text{ITT} / \text{compliance_rate}$$

Under SUTVA + randomization + monotonicity, this equals the ATE for compliers. Monotonicity means: no defiers (nobody takes treatment when assigned to control AND refuses treatment when assigned to treatment). This is usually a reasonable assumption.

APPENDIX B — COMMON STUDENT QUESTIONS

Q: Why do we need a t-test? Why not just compare the means directly?

A: Comparing means tells you the direction of the effect but not whether it is statistically distinguishable from zero. With small samples or noisy data, the observed difference could easily arise from random variation even with no true effect. The t-test formalizes the question: "How surprising would this difference be if the true effect were zero?"

Q: What is the difference between the p-value and the significance level alpha?

A: Alpha is the threshold you set before running the test — the maximum acceptable false positive rate. The p-value is the probability of observing data as extreme as yours under H_0 . You reject H_0 when $p < \alpha$. Alpha is your decision threshold; p is the test result.

Q: Can we ever prove that there is NO effect?

A: No — we can only fail to reject H_0 . The absence of significance is not evidence of absence of an effect. This is why equivalence tests (TOST — two one-sided t-tests) are used to demonstrate that an effect is smaller than some practical threshold. If you want to declare two variants equivalent, use TOST, not a standard t-test.

Q: What if our outcome is not normally distributed? Revenue data is very skewed.

A: The t-test relies on the normality of the SAMPLING DISTRIBUTION of the mean, not the outcome variable itself. By the Central Limit Theorem, the sample mean is approximately normal for large n, even if Y is highly skewed. With $n = 5,000$ per group, this is typically fine. For very small samples or extreme outliers, consider bootstrapped confidence intervals or non-parametric tests (Mann-Whitney U).

Q: Should we run a one-tailed or two-tailed test?

A: Default to two-tailed unless you have a strong prior that the effect can only go in one direction AND you are willing to commit before seeing data. Two-tailed is more conservative (harder to reject H_0) but avoids the problem of fishing for significance in one direction. Many A/B testing platforms use one-tailed by default — be aware of this when interpreting their output.

Q: What happens if we run the test for longer than planned?

A: This is "peeking" and it inflates the false positive rate. If you check significance every day for 30 days, the probability of a false positive is much higher than 5%. Solutions: pre-commit to a sample size before starting, use sequential testing methods (e.g., alpha-spending functions, mSPRT), or use a Bayesian testing framework that does not require pre-commitment.

Q: Can we A/B test things that are hard to randomize, like pricing?

A: Pricing is tricky because of market interference — if some customers see \$9 and others see \$12 for the same product on the same day, it can cause market distortions and customer complaints if discovered. Solutions: randomize by geography (different cities see different prices), randomize by time (week A vs week B), or use synthetic control methods.

Q: What is the difference between internal and external validity?

A: Internal validity: within your experiment, does the estimated effect correctly measure the causal effect for your study population? (Randomization provides this.) External validity: does the effect generalize to other populations, settings, or time periods? (Randomization does not guarantee this.) A/B tests on current platform users may not generalize to new user segments or different markets.

Q: Why does adding covariates to the regression reduce variance but not change the estimate?

A: Under randomization, treatment D is independent of all pre-treatment covariates X. So X does not explain any of the variation in D — adding X does not change the coefficient on D. But X does explain residual variation in Y — it absorbs variance that would otherwise be in epsilon, reducing the residual variance and therefore the SE of τ_{hat} . This is why collecting baseline data before the experiment is valuable.

Q: What is a novelty effect and how do we detect it?

A: Users respond to new experiences differently than they will after adaptation. A new UI might get higher engagement initially simply because it is new and different. To detect: run the experiment long enough for novelty to wear off, or compare effects across cohorts (early vs late users in the experiment window). Novelty effects typically decay over 2-4 weeks.

APPENDIX C — TIMING GUIDE

CLASS 1	Section	Minutes	Cumulative
	Three-act opening (Netflix/Amazon/Bing)	15	15
	Quick callback (PO → DiD → A/B)	5	20
	Why randomization works (math)	8	28
	Key vocabulary and estimands	5	33
	Worked example — 4 slides	20	53
	Power analysis and sample size	10	63
	Balance checks	5	68
	Notebook exercise (Your Turn)	20	88
	Debrief + wrap-up	5	93
	<i>Buffer / questions</i>	<i>~5</i>	<i>~75 target</i>

CLASS 2	Section	Minutes	Cumulative
	DIM = OLS and ANCOVA	7	7
	Non-compliance: ITT vs LATE	15	22
	Multiple testing and p-hacking	10	32
	Attrition bias	7	39
	Real case study discussion	8	47
	Statistical vs economic significance	5	52
	Break the Estimator — notebook exercise	25	77
	Bridge to text data + wrap-up	5	82
	<i>Buffer / questions</i>	<i>~5</i>	<i>~75 target</i>

Notes on pacing: The notebook exercises are the most likely to run long. If you are short on time, the "Break the Estimator" exercise (Class 2) can be assigned as homework with a debrief at the start of the next class. The three-act opening is the most likely to generate extended discussion — the Amazon example in particular tends to generate lots of UX-related tangents. Cut off discussion after 5 minutes per example and redirect to the punchline.